Applied Economics

OLS Variance and Robust Standard Errors

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Plan for this Lecture

1. Variance of the OLS Estimator
    - Special case of Homoskedasticity (MLR.5)
    - Role of Multicollinearity
2. Empirical Examples in R (robust standard errors)
3. Some more functionalities in R (ggplot2, writing functions, loops)

Variance of the OLS Estimator

Variance of OLS

To quantify uncertainty and for statistical inference we need to know/estimate the variance of $\hat{\boldsymbol{\beta}}$.

$$Var(\hat{\boldsymbol{\beta}}|\mathbf{X}) = \begin{pmatrix} Var(\hat{\beta}_0|\mathbf{X}) & cov(\hat{\beta}_0, \hat{\beta}_1|\mathbf{X}) & \dots & cov(\hat{\beta}_0, \hat{\beta}_k|\mathbf{X}) \\ cov(\hat{\beta}_0, \hat{\beta}_1|\mathbf{X}) & Var(\hat{\beta}_1|\mathbf{X}) & \dots & cov(\hat{\beta}_1, \hat{\beta}_k|\mathbf{X}) \\ \vdots & \vdots & \ddots & \vdots \\ cov(\hat{\beta}_0, \hat{\beta}_k|\mathbf{X}) & cov(\hat{\beta}_1, \hat{\beta}_k|\mathbf{X}) & \dots & Var(\hat{\beta}_k|\mathbf{X}) \end{pmatrix}$$

We are especially interested in the elements on the diagonal, which reflect the estimation uncertainty regarding the parameters $\beta_j, j = 0, 1, ..., k$

Variance of OLS

Remember, $\hat{\boldsymbol{\beta}}$ can be written as $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{u}$

Thus, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{u}$

This implies:

$$\begin{aligned} Var(\hat{\boldsymbol{\beta}}|\mathrm{X}) &= E[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})'|\mathrm{X}] \\ &= E[((\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{u})((\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{u})'|\mathrm{X}] \\ &= E[(\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\mathrm{u}\mathrm{u}'\mathrm{X}(\mathrm{X}'\mathrm{X})^{-1}|\mathrm{X}] \\ &= (\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'E[\mathrm{u}\mathrm{u}'|\mathrm{X}]\mathrm{X}(\mathrm{X}'\mathrm{X})^{-1} \end{aligned}$$

Thus, we need the variance matrix of the error terms $Var(\mathrm{u}|\mathrm{X}) = E[\mathrm{u}\mathrm{u}'|\mathrm{X}]$.

Variance of the error terms

$$Var(\mathrm{u}|\mathrm{X}) = E(\mathrm{uu}'|\mathrm{X}) = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix} = diag(\sigma_1^2, ..., \sigma_n^2) \equiv \Omega$$

The elements along the diagonal are the unknown variances of the error terms for all observations $i = 1, ..., n$. Why are the elements outside the diagonal all 0?

Variance of the error terms

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Since Ω is unknown, we need to estimate it. To do so, we replace σ_i^2 with the squared residual of observation i:

$$\widehat{\Omega} = diag(\hat{u}_1^2, \hat{u}_2^2, ..., \hat{u}_n^2)$$

Variance Estimator

Plugging $\widehat{\Omega}$ into the formula for $Var(\hat{\boldsymbol{\beta}}|\mathrm{X})$ yields a consistent estimator for the variance matrix of the OLS estimator:

$$\widehat{Var}(\hat{\boldsymbol{\beta}}|\mathrm{X}) = (\mathrm{X}'\mathrm{X})^{-1}\mathrm{X}'\hat{\Omega}\mathrm{X}(\mathrm{X}'\mathrm{X})^{-1}$$

Diagonal elements of this matrix: estimated variances of the (k+1) parameter estimators.
Taking the square roots yields the **standard errors** $\widehat{SE}(\hat{\beta}_j)$ which are relevant for hypothesis testing.

Heteroskedasticity vs. Homoskedasticity

The variance estimator we just derived is called **heteroskedasticity-robust** estimator. The corresponding standard errors are called heteroskedasticity-robust standard errors.

Per default, software programs like R report different standard errors that are based on a **homoskedasticity assumption (MLR.5)**.

MLR.5 Conditional Homoskedasticity

The errors u_i have constant conditional variance that does not depend on the values of the regressors: $var(u_i|\mathrm{X_i}) = \sigma_u^2$ for all $i = 1, ...n$.
In matrix notation: $var(\mathrm{u}|\mathrm{X}) = \sigma_u^2 I_n$,

Under this assumption the variance of the OLS estimator simplifies to:
$Var(\hat{\beta}|\mathrm{X}) = (\mathrm{X'X})^{-1}\sigma_u^2$

Heteroskedasticity vs. Homoskedasticity

MLR.5 is implausible in most economic applications.

Example: Among people with high levels of education, the variance in wages is larger than among people with low levels of education. $\rightarrow$ Unexplained variation in wages (Y) depends on education (X).

If MLR.5 is violated, the non-robust standard errors based on the simplified formula are inconsistent. The st. errors and t-statistics reported by standard software are invalid.

The robust standard errors are consistent under both conditional homoskedasticity and conditional heteroskedasticity. Typically, under MLR.5, they are a bit more conservative than necessary.

Heteroskedasticity-robust standard errors

Heteroskedasticity is viewed as the norm in most economic applications. Thus, the current empirical literature relies almost exclusively on robust standard errors.

In R, you can get the robust variance estimator using `vcovHC(model, type = "HC1")` from the package `sandwich`.

To report robust st. errors in the regression output, use `coeftest(model, vcov. = vcovHC(model, type = "HC1")` from the package `lmtest`.

Drivers of estimation uncertainty

We will assume homoskedasticity for a second, to analyse which factors drive the variance of the OLS estimator. (Also holds under heteroskedasticity, but is more difficult to show).

Variance of estimator $\hat{\beta}_j$ under homoskedasticity:

$$Var(\hat{\beta}_j|\mathrm{X}) = \frac{1}{n}\frac{\sigma_u^2}{\sigma_{X_j}^2}\frac{1}{1-R_j^2}$$

where R_j^2 is the R^2 from the regression of X_j on all other regressors X_l, $l \neq j$.

- shrinks as sample size n increases.
- shrinks as sample variation in regressor X_j increases.
- increases in the error variance σ_u^2.
- increases as the correlation between X_j and the other regressors increases..

Excursion: R^2 (goodness of fit)

R^2 is commonly reported in regression output in R, and measures what share of the total variation in the dependent variable is explained by variation in the regressors:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}, \text{ with}$$

Total sum of squares (SST): $\sum_{i=1}^{N}(y_i - \bar{y})^2$
Explained sum of squares (SSE): $\sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2$
Residual sum of squares (SSR): $\sum_{i=1}^{N} \hat{u}_i^2$

The (adjusted) R^2 is sometimes used to compare different models (for the same dependent variable Y), but it does not play a major role when your goal is causal analysis.

Multicollinearity and the OLS variance

R_j^2 is the R^2 from a regression of X_j on the other regressors.
$\rightarrow$ measures multicollinearity (linear correlation) between X_j and the other regressors.

- $0 \leq R_j^2 \leq 1$.
- $R_j^2 = 1$ ruled out by MLR.3 (no perfect multicollinearity).

Estimates are less precise (higher variance) if there is strong correlation among regressors.

- Intuition: If two regressors are strongly correlated, it is hard to disentangle their individual partial effects on Y.
- $\frac{1}{1-R_j^2}$: Variance Inflation Factor

Getting Robust standard errors in R

Non-robust St. Errors

«include=FALSE»= library(ggplot2) wage1 = read.csv("wage1.csv") @ «cache = F, message= F»= library(sandwich) for robust variance estimator library(lmtest) for regression tables

$reg_model < -lm(lwage\ educ + exper + I(exper^2) + female, data = wage1)coeftest(reg_model)useshomoskedasticerrors![sameassummary(reg_model)]@$

Robust St. Errors

«cache = F, message= F»= get the robust variance estimate

$\text{var}_r obust < -vcovHC(reg_m odel, type =$

$"HC1")regressiontablebasedontherobuststandarderrorscoeftest(reg_m odel, vcov. = var_r obust)@$

Creating Graphs with ggplot2

ggplot2 syntax

```
ggplot(data) + <GEOM FUNCTION>(aes(<MAPPINGS>))
```

Key components

- dataset containing the variables we want to plot
- one or more geom functions: the type of object we can observe in the plot, e.g. points, lines, boxplots, or bars.
- aesthetic attributes of the geometric object, e.g. x/y position, color, or shape. Aesthetic attributes are mapped to the variables in the dataset.

ggplot2 syntax

- geom_functions: geom_point, geom_smooth, geom_density, geom_col, ...
- in each geom functions, we have to specify the required aesthetic attributes, e.g. x- and y-variable, color, etc.
- to optimize the appearance of the plot we can change axes labels (`xlab("...")`, `ylab("...")`), choose a different theme (e.g. `theme_light()`, `theme_dark()`, `theme_minimal()`) or add a title (`ggtitle("...")`). These options are all added to ggplot(data) + geom_function(aes(...))

Using Loops in R

«cache = F, message= F,out.width='2.2in'»= for (i in 1:10) loop body: gets executed each time the value of i changes with each iteration @

The index does not have to be called `i`.

You cannot only loop through `1:10`, but also any other numbers collected in a vector.

Using Loops in R: Examples

«cache = F, message= F,out.width='2.2in'»=

for (j in c(1,3,5)) here we call the index j print(j) first print the current number print(j + 10) then add 10 to it and print the result @

Using Loops in R: Examples

«cache = F, message= F,out.width='2.2in'»= for (j in c(1,3,5)) here we call the index j and loop print(head(wage1[[j]])) through columns of a dataframe @

Recommended Literature

Heiß Chapters 1.9, 3.4, 8.1

Hanck et al. Chapter 5.4